



MID SEMESTER EXAMINATION, AUTUMN-2024
School of Applied Sciences, KIIT-DU

B. Tech.
1st Semester (2024AB)
Autumn 2024 (SAS)

Course: Physics	Code: PH10001	Full Marks: 20	Time: 90 minutes
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Answer any FOUR QUESTIONS including question No. 1 which is compulsory.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

All parts of a question should be answered at one place only.

Q.1		Marks	CO
a)	Compare the amplitude vs time graph for undamped and under-damped oscillators.	1	CO1
b)	Two pendulums, each of mass 100 gm are suspended from a rigid support by the weightless strings of length 50 cm each. The bobs of both the pendulums are connected with each other by a spring of force constant 2 N/m. Both the pendulums are subjected to in-phase mode of oscillation. What is the change in the time period of oscillation, if we replace the spring with another spring of force constant 4 N/m?	1	CO1
c)	Discuss the change in the maximum amplitude and resonant frequency, when the damping coefficient increases, in the case of a forced oscillator.	1	CO1
d)	A simple progressive wave is represented by the equation $y = 4 \sin \pi(4.0 t - 0.2x)$, Here x and y are in m and t is in second. Calculate the wavelength.	1	CO2
e)	Write any two conditions for sustained interference.	1	CO2

Q.2		Marks	CO
a)	Starting from the expression of amplitude for a steady state solution of a forced harmonic oscillator, derive the expression for the resonant frequency and the amplitude at resonance.	4	CO1
b)	The amplitude of a damped harmonic oscillator decays by 30% of the original value after 3 seconds. Calculate the damping coefficient of the medium.	1	CO1

Q.3		Marks	CO
a)	Establish the differential equation for a damped harmonic oscillator and find its general solution. Mention the condition under which the oscillator is overdamped.	4	CO1
b)	Identify the type of motion given by the following differential equation. $3 \frac{d^2x}{dt^2} + 12 \frac{dx}{dt} + 27x = 0$ Calculate the time period of oscillation.	1	CO1

Q.4		Marks	CO
a)	Set up the differential equation for a forced harmonic oscillator subjected to external periodic force with the angular frequency p . Also find the steady state solution.	4	CO1
b)	Differentiate between in-phase and out-of-phase mode of oscillations for a coupled oscillator.	1	CO1

Q.5		Marks	CO
a)	Define the interference of light. Explain different types of interference with examples and necessary diagrams.	4	CO2
b)	Write down the differential equation of wave motion, for a progressive wave propagating along positive x -axis, and define the meaning of the symbols used.	1	CO2
